

Lecture 17: Three Learning Principles

Caltech CS156 Technical Study Guide

Gemini Notebook

September 18, 2026

Executive Summary

Machine learning application relies on three fundamental overarching principles: Occam's Razor (simplicity in modeling), Sampling Bias (distribution alignment), and Data Snooping (data integrity). Violating any of these principles leads to severe generalization failures regardless of mathematical sophistication.

The Three Foundational Principles of Machine Learning

Occam's Razor

Sampling Bias

Data Snooping

Simpler model is most plausible (falsifiability & counting argument)
 $P(\text{train})$ must equal $P(\text{test})$ (Dewey vs Truman 1948 poll)
Data affects choices -> charge dVC!
(never normalize test set with train)

Figure 1: The Three Foundational Principles governing sound machine learning practice.

1 Principle 1: Occam's Razor

Definition: "The simplest model that fits the data is also the most plausible."

1.1 Object vs. Set Complexity

- **Object Complexity:** Intrinsic description length of a single hypothesis h (e.g., polynomial degree or minimum description length in bits).
- **Set Complexity:** Diversity of the hypothesis space $\mathcal{H}$ (e.g., d_{VC} or growth function $m_{\mathcal{H}}(N)$).

The Counting Link: A hypothesis is simple if it belongs to a small hypothesis set $|\mathcal{H}|$. Because $|\mathcal{H}|$ is small, fitting a given dataset by chance is extremely unlikely ($m_{\mathcal{H}}(N)/2^N \rightarrow 0$). When a simple model does fit, the fit is meaningful and provides high generalization guarantees.

1.2 Falsifiability

An explanation is scientific only if it is falsifiable. If a dataset of $N = 2$ points is fit by a line ($d_{VC} = 2$), no evidence is gained because 2 points can always be connected by a line ($m_{\mathcal{H}}(2) = 2^2$). Falsifiability requires $N > d_{VC}$.

2 Principle 2: Sampling Bias

Definition: *"If the data is sampled in a biased way, learning will produce a biased outcome."*

2.1 Distributional Misalignment

The core theoretical assumption in machine learning is $P_{\text{train}}(\mathbf{x}) = P_{\text{test}}(\mathbf{x}) = P(\mathbf{x})$. When $P_{\text{train}} \neq P_{\text{test}}$, E_{in} is no longer a valid proxy for E_{out} .

Historical Example (Dewey Defeats Truman, 1948): A telephone poll predicted Dewey's victory over Truman. Telephones in 1948 were luxury items, causing severe sampling bias towards wealthier voters who favored Dewey.

3 Principle 3: Data Snooping

Definition: *"If a dataset has affected any learning choice, its ability to assess performance is compromised."*

3.1 Manifestations of Data Snooping

1. **Visual Inspection:** Looking at scatter plots to choose non-linear feature transforms before training (d_{VC} must charge for the human inspection step).
2. **Global Preprocessing:** Calculating mean μ and variance σ^2 across the *entire* dataset prior to splitting into train/test sets.
3. **Reusing Test Sets:** Iteratively tuning hyperparameters against a test set converts it into a validation set with $M > 1$ selection bias.